PNAS RESEARCH ARTICLE | NEUROSCIENCE

Secondary thalamic dysfunction underlies abnormal large-scale neural dynamics in chronic stroke

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Edited by Marlene Behrmann, University of Pittsburgh, Pittsburgh, PA; received May 10, 2024; accepted September 18, 2024

Stroke causes pronounced and widespread slowing of neural activity. Despite decades of work exploring these abnormal neural dynamics and their associated functional impairments, their causes remain largely unclear. To close this gap in understanding, we applied a neurophysiological corticothalamic circuit model to simulate magnetoencephalography (MEG) power spectra recorded from chronic stroke patients. Comparing model-estimated physiological parameters to those of controls, patients demonstrated significantly lower intrathalamic inhibition in the lesioned hemisphere, despite the absence of direct damage to the thalamus itself. We hypothesized that this disinhibition could instead be related to secondary degeneration of the thalamus, for which growing evidence exists in the literature. Further analyses confirmed that spectral slowing correlated significantly with overall secondary degeneration of the ipsilesional thalamus, encompassing decreased thalamic volume, altered tissue microstructure, and decreased blood flow. Crucially, this relationship was mediated by model-estimated thalamic disinhibition, suggesting a causal link between secondary thalamic degeneration and abnormal brain dynamics via thalamic disinhibition. Finally, thalamic degeneration was correlated significantly with poorer cognitive and language outcomes, but not lesion volume, reinforcing that thalamus damage may account for additional individual variability in poststroke disability. Overall, our findings indicate that the frequently observed poststroke slowing reflects a disruption of corticothalamic circuit dynamics due to secondary thalamic dysfunction, and highlights the thalamus as an important target for understanding and potentially treating poststroke brain dysfunction.

stroke | thalamus | magnetoencephalography | computational modeling | aperiodic neural dynamics

Significance

It is not currently understood why stroke causes a prominent slowing of neural activity, nor the associated functional impairment. Using a computational corticothalamic circuit model, we find that thalamic dysfunction is central to this abnormal activity as measured by magnetoencephalography (MEG), despite the absence of direct damage to the thalamus itself. The model indicated that slowing was caused by thalamic disinhibition, which was in turn associated with secondary thalamus degeneration. These findings highlight secondary thalamus disruption as an important determinant of corticothalamic circuit dysfunction and poststroke patient outcomes, and demonstrate how noninvasive measures of brain activity like MEG can be linked to underlying stroke pathophysiology via computational modeling.

Stroke is among the leading causes of disability worldwide (1), leaving millions of individuals with lasting motor, cognitive, and linguistic impairments each year. Given the growing burden that stroke-related disability poses as global demographics change (2), there is a critical need for new interventions that directly modify brain function to improve recovery, as well as means of tailoring new and existing therapies to treat individuals most effectively. However, these goals require biomarkers that are well understood in terms of the disease processes they reflect. Although there is a long history of using electroencephalography (EEG), and more recently magnetoencephalography (MEG), to quantify and diagnose brain dysfunction in stroke (3–9), there remains little understanding of how stroke pathophysiology causes the abnormal neural dynamics captured by these techniques.

Computational models of neural circuits can be used to bridge the gap in understanding between pathophysiology and noninvasive measures of brain dynamics. They achieve this by simulating the interaction between mathematically abstracted populations of neurons, the behavior of which is dictated by a small number of physiologically inspired parameters. In doing so, these models can produce realistic emulations of large-scale neural activity of the kind captured by M/EEG, while also remaining computationally tractable. By fitting this simulated activity to empirical M/EEG data, these models can be used to estimate otherwise inaccessible physiological quantities, and thereby offer insight into the causes of abnormal large-scale neural dynamics.

Accomplishing this requires a physiological model that can reproduce the abnormal dynamics observed in stroke. Across stroke electrophysiology, the dominant finding historically has been abnormally high power at lower frequencies (typically delta/theta) in both acute and chronic patients (7, 9–14), and such enhanced low-frequency power has been associated with poorer outcomes for patients (7, 13, 15–17). However, we recently demonstrated in chronic stroke patients that this abnormal elevation in low-frequency power was better explained by a steepening of the aperiodic component of the power

Author contributions: P.R.J., J.D.G., A.R.M., and J.A.M. designed research; P.R.J. and J.A.M. performed research; P.R.J. and L.R. analyzed data; P.R.J. performed visualization; L.R. performed data and code validation; A.R.M. provided supervision; J.A.M. provided data; J.D.G., L.R., A.R.M., and J.A.M. edited the paper; and P.R.J. wrote the paper.

The authors declare no competing interest.

This article is a PNAS Direct Submission.

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This article contains supporting information online at https://www.pnas.org/lookup/suppl/doi:10.1073/pnas.2409345121/-/DCSupplemental.

Published November 6, 2024.

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spectrum, rather than an enhancement of low-frequency oscillations (18), and work in rats has reached similar conclusions (19). This steepening reflects a larger exponent (N) in the function $P \propto 1/f^N$, which defines the exponential falloff in M/EEG signal power P as a function of frequency f (20–22). However, while elevated low frequency power was not attributable to low frequency oscillations, we did observe abnormal oscillatory dynamics at higher frequencies, namely slowing of the alpha oscillation and diminished beta oscillation power (18).

Given the need to simulate both aperiodic and periodic (oscillatory) activity to capture these abnormal dynamics in stroke, we chose the corticothalamic model developed by Robinson and colleagues (23, 24), which simulates the interaction between cortical and thalamic neuronal populations. Crucially, this model accurately captures both the aperiodic and periodic components of the M/EEG power spectrum, and has been successful in reproducing a variety of macroscale electrophysiological phenomena detectable with EEG in healthy and clinical populations (23–28).

Regarding the underlying pathophysiology causing abnormal dynamics, current evidence suggests that both noninfarcted cortex and thalamus are plausibly disrupted in stroke (29–32). Applying this corticothalamic model could, therefore, clarify how dysfunction in either the cortex or thalamus contributes to the generation of abnormal neural dynamics within the larger corticothalamic circuit. In the cortex, for instance, stroke is known to produce a variety of adverse effects, including increased tonic GABAergic inhibition (29) in the noninfarcted cortex peripheral to the lesion, as well as thinning of distant cortex connected to the lesioned area (30).

However, there is also growing evidence that the thalamus is particularly susceptible to degeneration, even when the infarction occurs elsewhere in the brain. Converging evidence that some patients experience such "secondary" thalamic degeneration, particularly of the ipsilesional thalamus, includes: reduced thalamic volume (31–37), increased mean diffusivity (MD) suggesting abnormal tissue microstructure (34, 38–41), loss of central benzodiazepine receptors indicating neuronal loss (42), decreased perfusion (42–44) and decreased metabolic rate (42, 43, 45, 46). Similarly, loss of thalamic neurons (47) and myelin (48) has been demonstrated in rodents following middle cerebral artery occlusion.

Given that the thalamus is involved in cognition, language, and motor function (49, 50), such degeneration would be expected to have important implications for stroke recovery. Recent evidence has indeed related poststroke cognitive impairment to ipsilesional thalamic atrophy (32), neuronal loss and decreased metabolic rate (42), and related motor impairment to thalamic atrophy (31).

Considering this evidence, our aim in the present work was to use mathematical modeling of corticothalamic neural dynamics, in combination with various markers of tissue pathology, to unify understanding of stroke pathophysiology, the abnormal brain dynamics indexed by M/EEG, and cognitive outcomes. To achieve this, we fitted simulated power spectra from the corticothalamic model to empirical resting-state MEG power spectra from 18 chronic aphasic stroke patients and 23 age- and education-matched controls. The patients' model-derived parameters were then related to underlying brain pathology as derived from structural MRI, diffusion-weighted imaging (DWI), and arterial spin labeling (ASL), as well as a limited number of cognitive and language measures.

1. Results

1.1. Corticothalamic Circuit Model Accurately Captures Subject- and Region-Specific Variations in Resting-State MEG Power Spectra. To identify underlying neurophysiological abnormalities associated with stroke-related MEG features, we fit a neurophysiological model of the corticothalamic system (23, 51) to empirical MEG power spectra (Fig. 1) from 18 chronic stroke patients with a single left-hemispheric stroke (Table 3), and 23 age- and education-matched controls. This model has been successful in reproducing a variety of electrophysiological phenomena (23, 25–27), particularly in the lower end of the power spectrum, which is dominated by aperiodic activity and alpha oscillations—two features that are central to the abnormal brain dynamics observed after stroke (18). Activity in corticothalamic circuits is represented using four coupled neuronal populations: excitatory cortical neurons (e), inhibitory cortical neurons (i), neurons in the excitatory relay nuclei of the thalamus (s), and neurons in the inhibitory reticular nucleus of the thalamus (r). In the context of steady-state system behavior, as is the case for the resting-state MEG data studied here, the second-order differential equations describing each neuronal population's membrane potential and firing rate dynamics can be simplified to a linear transfer function form in the frequency domain. These linearized model equations (*SI Appendix*, section S.2) allow direct calculation of an analytic power spectrum from physiological circuit parameters across the cortical and thalamic populations, as well as efficient estimation of these parameters from M/EEG data.

For each participant, model-generated power spectra were fit to resting-state empirical power spectra from virtual channels centered on the 78 cortical regions of the Automated Anatomical Labeling (AAL) atlas (52). Specifically, a Markov Chain Monte Carlo (MCMC) method (51) was used to sample the joint probability distribution of model parameters and identify the set that maximizes the likelihood estimate $L = e^{-\frac{\chi^2}{2}}$ (bounded between 0 and 1), where χ^2 is a weighted measure of the difference between model-generated and empirical power spectra (see Section 3.4 and *SI Appendix* for further details).

Model fits were good overall, with a mean likelihood estimate of 0.75 (SD = 0.07) for patients and 0.73 (SD = 0.07) for controls. Likelihood estimates for the ipsilesional (left) hemisphere were better for patients on average (mean = 0.76, SD = 0.06) than controls (mean = 0.72, SD = 0.07; $t(39) = 2.11$, $P = 0.041$, $d = 0.66$, 95% CI [0.003, 0.083]). There was no significant difference in likelihood estimates in the contralesional (right) hemisphere between patients (mean = 0.74, SD = 0.07) and controls (mean = 0.73, SD = 0.06; $t(39) = 0.42$, $P = 0.676$, $d = 0.13$, 95% CI [−0.03, 0.05]). Examples of empirical and fitted power spectra are shown in Fig. 1*D*, demonstrating the model's ability to parsimoniously capture important aspects of the MEG power spectrum, particularly the low-frequency 1/f component and alpha peak.

1.2. Stroke Patients Show Decreased Intrathalamic Inhibition in the Ipsilesional Hemisphere. The corticothalamic model described above and in Fig. 1*A* has nine fitted parameters (see *SI Appendix*, Table S1 for full descriptions), defining synaptic weights, kinetics, and conduction delays within the e, i, r, and s populations, and collectively determining their dynamics and corresponding simulated MEG spectra. Of primary interest were those based on the "gain" parameters G_{ab}, which reflect the average synaptic weight or strength of the connection from population a to population b. The five fitted gain parameters (illustrated in Fig. 1*A*) were G_{ee}, G_{ei}, and the three composite gains $G_{ese} = G_{es}G_{se}$, $G_{esre} = G_{es}G_{sr}G_{re}$ and $G_{srs} = G_{sr}G_{rs}$. The remaining four fitted parameters were the synaptic decay (α) and rise (β) times, the time delay in thalamocortical transmission (t_0), and a correction factor for high frequency muscle artifact (A_{EMG}). Finally, the main dynamics of the model in relation to the above parameters can be represented in a simplified form with three aggregate parameters X, Y, and Z

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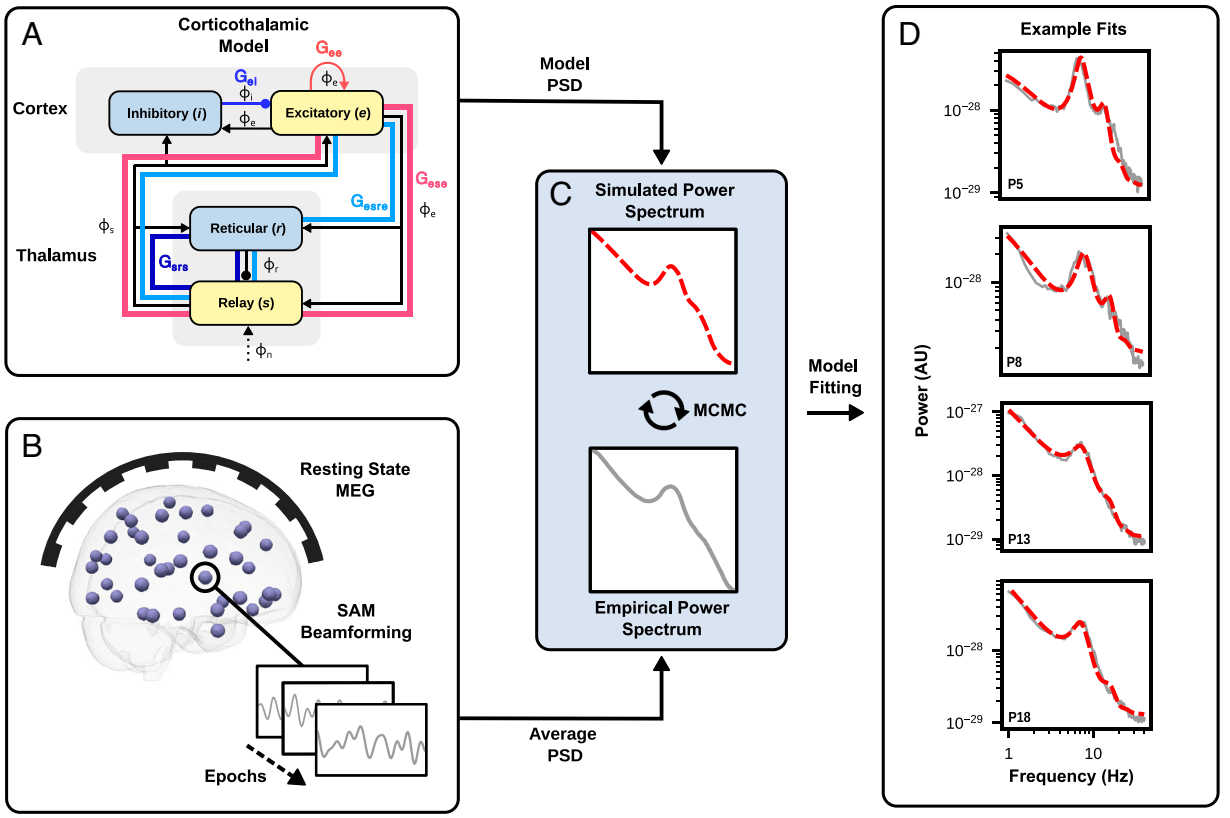


Fig. 1. Model fitting methodology. (*A*) The neurophysiological model of the corticothalamic system used to simulate power spectra. This model emulates the interactions among two cortical neuronal populations (one inhibitory [*i*] and one excitatory [*e*]) and two thalamic neuronal populations (the excitatory relay nuclei [*s*] and inhibitory reticular nucleus [*r*]). Excitatory populations are represented by yellow boxes, and inhibitory populations by blue boxes. This model is described by five gain parameters (G_{ee}, G_{ei}, G_{ese}, G_{esre}, G_{srs}) representing the ratio of input to output within key loops of the circuit, as well as two parameters shared by all four populations (α: synaptic decay time, β: synaptic rise time) and two additional parameters (t_0: thalamocortical delay, A_{EMG}: electromyographic artifact correction). Φ symbols represent the axonal signals propagating from their respective neuronal populations, with the simulated power spectrum being derived from Φ_e. These parameters can be further aggregated into three simplified parameters *X, Y,* and *Z*, which summarize the gain of the three major loops: intracortical, corticothalamic, and intrathalamic, respectively (see Section 3.4 and *SI Appendix*). (*B*) Empirical power spectra were derived from resting state MEG recordings, source-localized with synthetic aperture magnetometry (SAM) beamforming. Time series were computed for spheres of 10 mm diameter centered on the 78 cortical regions of the Automated Anatomical Labeling (AAL) atlas. Power spectral densities (PSDs) were computed on 5 s epochs using Welch's method, then averaged over all epochs for each sphere. (C) Corticothalamic model parameters were estimated using BrainTrak, which employs a Markov Chain Monte Carlo (MCMC) method to sample the joint probability distribution of the nine model parameters, seeking the set of parameters that minimizes the difference between simulated and empirical power spectra (see Section 3.4 and *SI Appendix* for further details). (*D*) Examples of fitted power spectra from patients (dotted red lines) compared to their corresponding empirical power spectra (solid gray lines). Selected power spectra are from the left (ipsilesional) postcentral gyrus for all four patients.

(see equations in Section 3.4), which respectively parameterize the overall intracortical, thalamocortical, and intrathalamic loop gains of the circuit (53).

Having confirmed the accuracy of model fits across participants and regions (Section 1.1), we first compared estimated values of the three aggregate loop gain parameters between patients, controls, and hemispheres. To account for the large heterogeneity in lesion size and location, parameters were averaged over all 39 AAL regions within each of the left and right hemispheres. From these comparisons, only *Z* (representing the net intrathalamic gain) was significantly diminished in the ipsilesional hemisphere compared to controls (Fig. 2*C*; $t(39) = -5.66$, $P < 0.001$, $d = -1.78$, 95% CI [−0.10, −0.05]). No statistically significant difference between patients and controls was detected in *X*, representing intracortical gain ($t(39) = 1.26$, $P = 0.216$, $d = 0.4$, 95% CI [−0.01, 0.06]), nor in *Y*, representing thalamocortical gain (t(39) = 0.59, $P = 0.56$, $d = 0.19$, 95% CI [−0.015, 0.027]). Fig. 2*D* illustrates the effect of decreasing *Z* while holding all other parameters constant, causing a pattern of changes in the power spectrum similar to that observed in patients compared to controls (Fig. 2*B*)—namely greater low-frequency power, reduced high-frequency power, and slower alpha rhythm peak frequency.

Among the nine lower-level parameters from which *X, Y,* and *Z* are derived, G_{srs} (the intrathalamic inhibition gain underlying *Z*) showed by far the greatest intergroup difference ($t(39) = 5.59$, $P < 0.001$, $d = 1.76$, 95% CI [0.58, 1.22]). G_{ese} ($t(39) = -2.70$, $P = 0.010$, $d = -0.85$, 95% CI [−4.26, −0.66]) and A_{EMG} ($t(39) = 2.53$, $P = 0.015$, $d = 0.8$, 95% CI [0.01, 0.05]) also showed notable differences, although these did not survive Bonferroni correction. Since hemisphere-averaged *Z* and G_{srs} were highly anticorrelated among patients ($r(16) = -0.9$, $P \ll 0.001$, 95% CI [−0.96, −0.75]), we conclude that elevated (less negative) G_{srs} drives the group differences in Z described above. In the contralesional (right) hemisphere, significantly lower *Z* ($t(39) = -3.13$, $P = 0.003$, $d = -0.99$, 95% CI [−0.07, −0.02]) and higher G_{srs} ($t(39) = 3.30$, $P = 0.002$, $d = 1.04$, 95% CI [0.21, 0.82]) were seen in the patient group compared to controls.

1.3. Stroke Patients Show Ipsilesional Thalamic Degeneration Across Multiple Imaging Measures. After manual thalamus segmentation following a published protocol (54), stroke patients showed evidence of thalamic degeneration on all available MRI measures (Fig. 3*A*). Specifically, patients showed significantly smaller ipsilesional thalamus volumes ($t(39) = -5.02$, $P < 0.001$,

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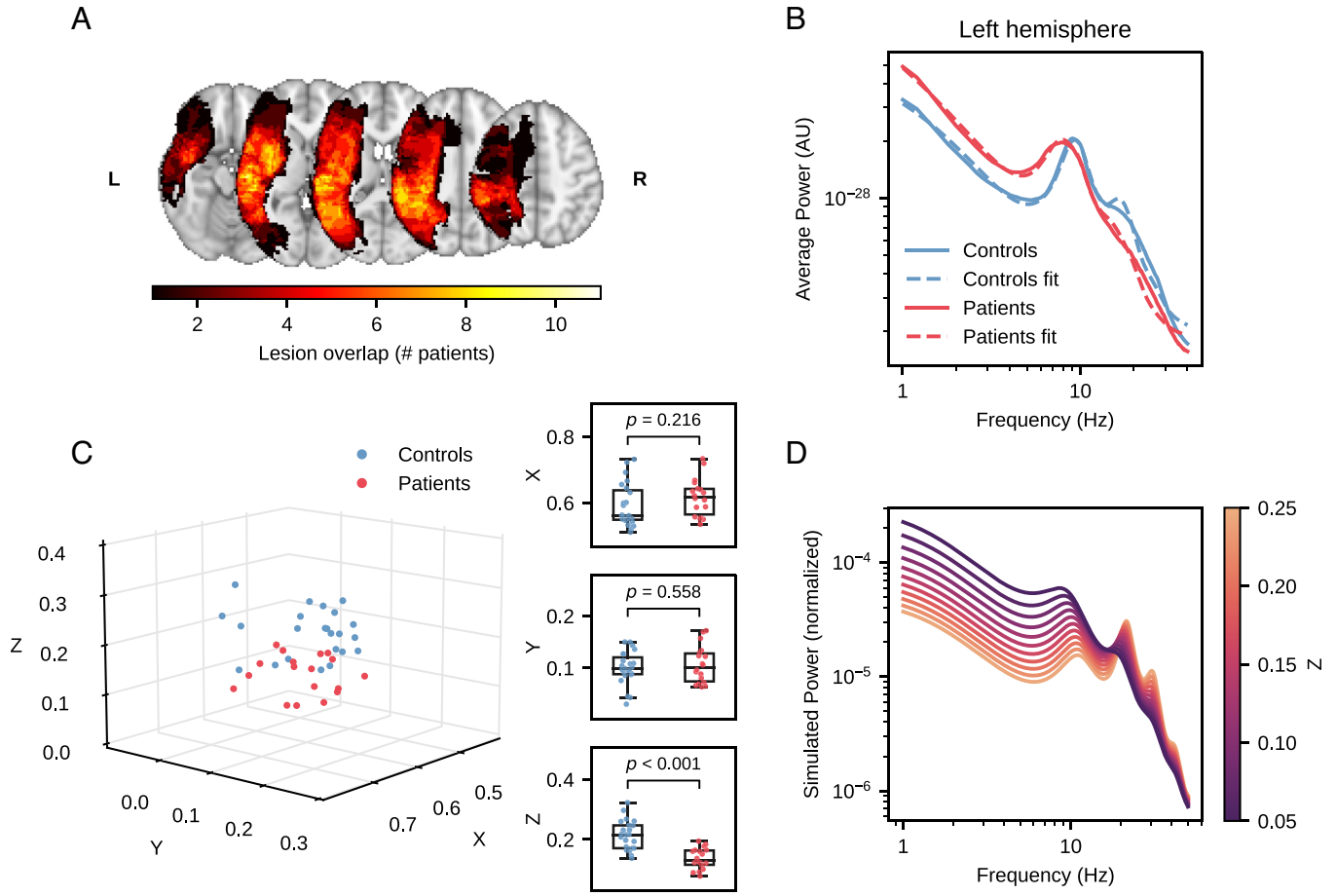


Fig. 2. Reduced ipsilesional thalamic inhibition in chronic stroke. Corticothalamic modeling analyses revealed a reduction of intrathalamic inhibitory feedback (reduced model parameter *Z*) in chronic stroke patients, providing a mechanistic explanation for the observed spectral slowing in the ipsilesional hemisphere. (*A*) All patients had a lesion in the left hemisphere, with the highest concentration of lesions affecting superior temporal gyrus, Heschl's gyrus, and supramarginal gyrus (median 41.2%, 24.75%, and 19.4% lesioned by volume, respectively). (*B*) On average, patients exhibit "spectral slowing" compared to controls, involving a higher aperiodic exponent, alpha slowing, and decreased beta power. (*C*) Patients exhibit significantly lower average left hemisphere fitted *Z* parameter compared to controls, representing reduced intrathalamic inhibition in the corticothalamic model. Center lines on the box plots represent the median, boxes represent the interquartile range (IQR), and whiskers represent 1.5 times the IQR. (*D*) When all other model parameters are held constant ($X = 0.41$, $Y = 0.51$, $\alpha = 83.33$, $\beta = 769.23$, $t_0 = 0.085$) lower *Z* values produce a similar slowing pattern in fully simulated spectra as that observed in patients. This includes increased low frequency power, decreased alpha frequency, and decreased power above 15 Hz as *Z* decreases.

$d = -1.58$, 95% CI [−2.91, −1.27]) and higher mean diffusivity (MD; $t(38) = 3.97$, $P < 0.001$, $d = 1.27$, 95% CI [1.02, 3.02]) compared to controls. Patients also showed a nonsignificant reduction in ipsilesional thalamic cerebral blood flow (CBF; $t(37) = -0.93$, $P = 0.36$, $d = -0.3$, 95% CI [−1.38, 0.50]) compared to controls.

Importantly, principal components analysis (PCA) revealed substantial shared variance among ipsilesional thalamus volume, MD, and CBF estimates, potentially indicating a common underlying degeneration process affecting all three (Fig. 3*C*). The first principal component accounted for 62.6% of the shared variance among the three measures in patients. All three measures loaded on this component with a similar magnitude, and each in the direction associated with thalamic degeneration (volume: −0.51, MD: 0.60, CBF: −0.62). As such, an individual patient's projection onto this principal component was considered their "thalamic degeneration score," and was used to quantify overall thalamic degeneration for subsequent analyses.

Interestingly, we observed evidence for the opposite effect in the contralesional hemisphere, with patients showing larger right thalamus volumes than controls ($t(39) = 2.04$, $P = 0.048$, $d = 0.64$, 95% CI [0.02, 1.23]). No significant group differences were observed for right hemisphere thalamic MD ($t(38) = 0.79$, $P = 0.434$, $d = 0.25$, 95% CI [−0.45, 1.06]) or CBF ($t(37) = -0.04$, $P = 0.966$, $d = -0.01$, 95% CI [−0.99, 0.94]), however.

1.4. Model-Estimated Intrathalamic Inhibition Links Thalamic Degeneration and Abnormal MEG Dynamics. Patients' ipsilesional thalamus degeneration scores were significantly correlated with both their average ipsilesional *Z* parameter ($\beta = -0.55$, $P = 0.006$, 95% CI [−0.81, −0.11]) and an ipsilesional MEG slowing score ($\beta = 0.46$, $P = 0.035$, 95% CI [−0.01, 0.76]; Fig. 4*A*) computed on spectral parameters extracted from their power spectra (*Section 3.3*). PCA computed on these measures yielded a first principal component accounting for 75.7% of their shared variance (Fig. 4*B*). Loadings on this component indicated that greater thalamus degeneration is related to lower *Z* and more MEG slowing (thalamus degeneration score: 0.49, average *Z* parameter: −0.63, MEG slowing score: 0.61). Examples of patients with high and low scores on this principal component are shown in Fig. 5.

Lesion volume was not significantly correlated with ipsilesional thalamus degeneration ($\beta = 0.08$, $P = 0.73$, 95% CI [−0.40, 0.53]), average *Z* ($\beta = 0.16$, $P = 0.50$, 95% CI [−0.33, 0.59]), or MEG slowing ($\beta = -0.09$, $P = 0.72$, 95% CI [−0.53, 0.39]), indicating that the shared variance among these measures was not merely a function of stroke severity.

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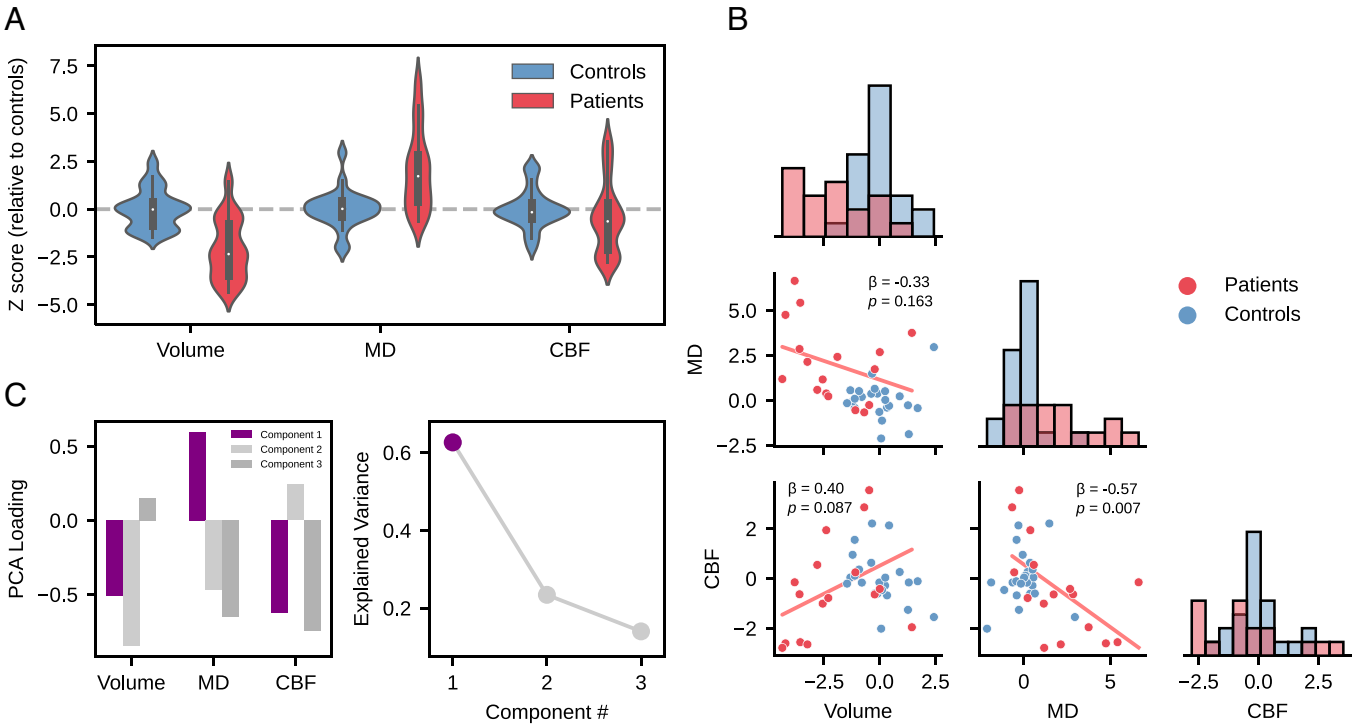


Fig. 3. MRI markers of stroke-related thalamic degeneration. Patients show converging evidence of ipsilesional thalamic degeneration across multiple MRI measures, and a single "thalamus degeneration score" can be computed for each patient. (*A*) Compared to controls, patients show decreased left (ipsilesional) thalamus volume, increased MD, and decreased CBF (n.s.). All values have been normalized by subtracting the mean and dividing by the SD of the controls, and the dotted gray line represents the mean of the control group after normalization (zero). White dots on the violin plots represent the median, boxes represent the IQR, and whiskers represent 1.5 times the IQR. (*B*) Across patients, the pairwise relationships between these variables were in the expected directions, with the relationship between MD and CBF passing a significance threshold of $P < 0.05$. Histograms show the distribution of the variable labeled at the bottom of the column. (*C*) The correlation between these variables is well explained by a single principal component (*Left*, in purple) which accounts for 62.6% of the variance (*Right*). A patient's score on this principal component was considered their thalamus degeneration score for subsequent analyses.

Finally, we employed Causal Mediation Analysis (Mediation v5.4.0) (55) to test the following hypothesized causal explanation for abnormal MEG dynamics observed in stroke patients (Fig. 4*C*): secondary thalamic degeneration is accompanied by disinhibition of the thalamus, which in turn produces abnormal neural dynamics that manifest as MEG spectral slowing. Consistent with this hypothesis, the total effect of thalamus degeneration score on MEG slowing score (c = 0.46, 95% CI [0.09, 0.85], $P = 0.012$) was completely mediated by the indirect effect of the Z parameter (ab = 0.49, 95% CI [0.19, 0.91], $P = 0.002$), leaving the remaining direct effect nonsignificant (c′ = −0.04, 95% CI [−0.30, 0.17], $P = 0.654$). Subsequent sensitivity analysis indicated that this effect is robust to unmeasured mediator-outcome confounding ($\rho_{ACME=0} = -0.8$).

1.5. Proximity to Lesion Predicts MEG Spectral Slowing. Having linked thalamus degeneration to generalized slowing in the hemisphere-averaged MEG power spectra, we also considered how MEG power spectra varied regionally. Specifically, we considered whether a region's proximity to the lesion predicted abnormal slowing, over and above that predicted by thalamus degeneration. We addressed this question with a linear mixed effects model (lme4 v1.1-34) (56), predicting regional MEG slowing scores as a function of the following fixed effects: ipsilesional thalamus degeneration (one value per patient), proximity to lesion (one value per region × patient), lesion volume (one value per patient), and the interaction between ipsilesional thalamus degeneration score and proximity to lesion. We included a random intercept and slope (for proximity to lesion) for each patient, and also included the interaction between lesion proximity and thalamus degeneration. A region's proximity to the lesion was defined as the Euclidean distance between the center of the virtual MEG channel and the nearest voxel in the lesion mask.

Mixed effects model results are summarized in Table 1. Proximity to lesion significantly predicted regional MEG slowing score (estimate = 0.36, SE = 0.08, $t = 4.74$, $P < 0.001$). The model suggested an effect of overall thalamus degeneration on regional MEG slowing (estimate = 0.27, SE = 0.15, $t = 1.81$, $P = 0.09$), although it did not pass the predetermined significance threshold of $\alpha = 0.05$. Lesion volume was not a significant predictor of regional MEG slowing (estimate = −0.16, SE = 0.14, $t = -1.125$, $P = 0.278$).

1.6. Thalamic Degeneration Is Correlated with Cognitive and Language Impairment. Finally, we considered whether ipsilesional secondary thalamic degeneration predicted general cognitive outcomes, as indexed by the Montreal Cognitive Assessment (MOCA; n = 13), and aphasia symptoms as indexed by the Boston Naming Test (BNT; n = 17), Peabody Picture Vocabulary Test (PPVT; n = 17), and Western Aphasia Battery (WAB; n = 17). All variables were first residualized by lesion volume and standardized to control for the possible shared effect of stroke severity on thalamus degeneration and behavioral outcomes. Ipsilesional thalamus degeneration score was significantly related to outcomes (Fig. 6) on the MOCA ($r_s(11) = -0.69$, $P = 0.01$), the BNT ($r_s(15) = -0.56$, $P = 0.019$), and PPVT ($r_s(15) = -0.56$, $P = 0.02$). Correlation with the WAB was negative as expected, but did not pass the significance threshold ($r_s(15) = -0.29$, $P = 0.256$).

We also considered the differential effects of thalamus degeneration and primary cortical tissue damage on aphasia symptoms (Table 2). Specifically, we tested multiple linear regression models predicting the three aphasia measures (BNT, PPVT, and WAB)

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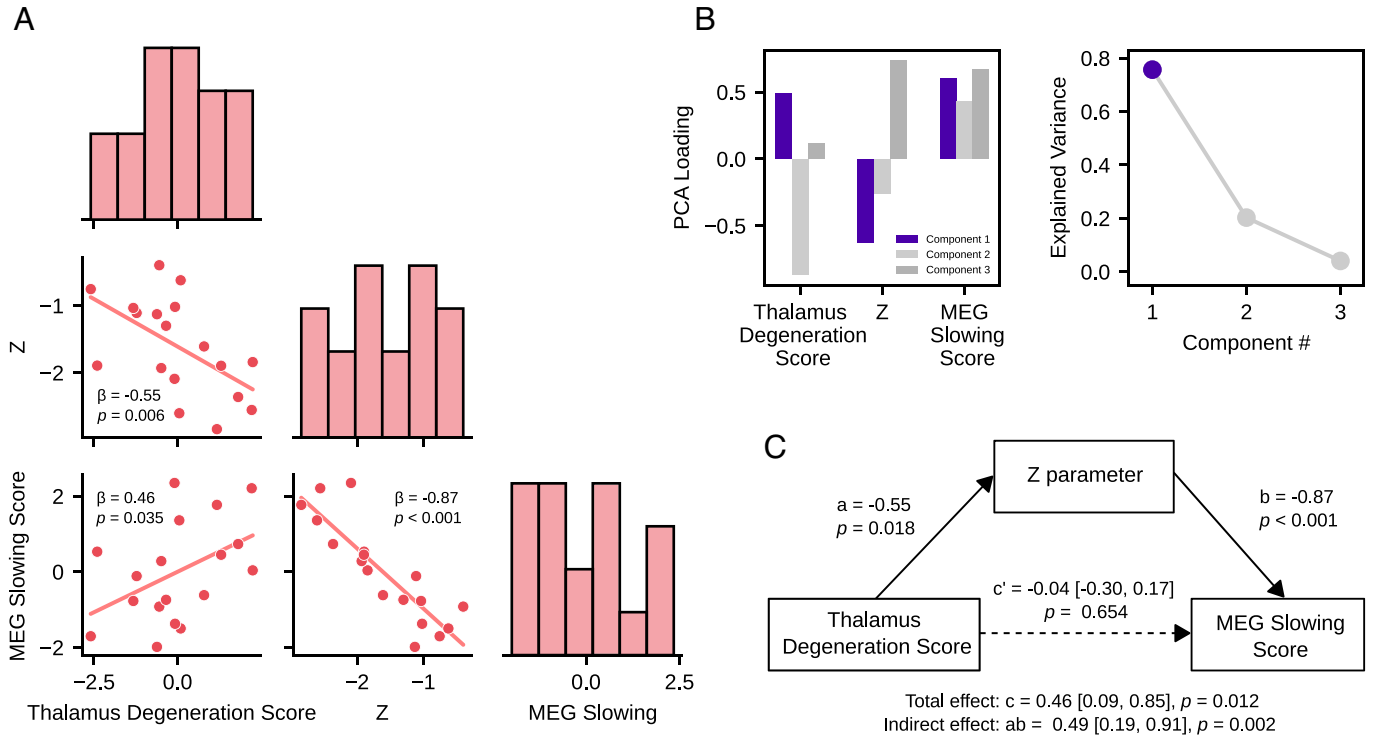


Fig. 4. Model-estimated intrathalamic inhibition links thalamic degeneration and abnormal MEG dynamics. In the ipsilesional (left) hemisphere, MEG slowing score was predicted by thalamus degeneration score, and mediated by the fitted *Z* parameter, suggesting that MEG slowing could be the result of thalamic disinhibition related to secondary thalamic degeneration. (*A*) Patients show significant pairwise correlations between average ipsilesional thalamus degeneration, fitted *Z* parameter, and MEG slowing score. Histograms show the distribution of the variable labeled at the bottom of the column. (*B*) Across patients, the correlation between these variables is well explained by a single principal component (*Left*, in dark purple) which accounts for 75.7% of the shared variance (*Right*). (*C*) The effect of ipsilesional thalamus degeneration score on ipsilesional MEG slowing score is mediated by the average ipsilesional fitted *Z* parameter, representing thalamic disinhibition. Solid paths are significant at α = 0.05, while the dashed path is not significant. Square brackets contain 95% CIs estimated with bootstrap resampling (1,000 resamples).

from thalamus degeneration score and the proportion of damage to classical language areas (left superior temporal gyrus, left angular gyrus, and left inferior frontal gyrus). Thalamus degeneration was the best predictor of BNT (β = −0.55, P = 0.035) and PPVT (β = −0.51, P = 0.047), while the WAB was better predicted by damage to the left inferior frontal gyrus (β = −0.90, P = 0.003).

2. Discussion

Taken together, our findings suggest that secondary damage of the ipsilesional thalamus in chronic stroke patients causes generalized disruption to corticothalamic circuit dynamics, manifesting as altered MEG power spectra in the ipsilesional hemisphere, and possibly leading to impaired cognition and language. These results add to the growing body of evidence showing that some individuals experience secondary degeneration of the ipsilesional thalamus after a stroke elsewhere in the brain, and that this degeneration is related to poorer outcomes (31, 32, 42). Importantly, we have also demonstrated a correspondence between reduced volume, increased MD, and decreased CBF in the thalamus—three measures of thalamic degeneration which have yet to be directly compared in the literature to our knowledge. Moreover, we used a physiological model to demonstrate that the relationship between thalamic degeneration and MEG slowing is mediated by thalamic disinhibition (Z), putatively resulting from decreased input from the inhibitory thalamic reticular nucleus (G_{sys}), rather than abnormalities in intracortical gain (X) or thalamocortical gain (Y). We therefore propose that the widespread slowing of brain dynamics often detected with M/EEG in stroke is primarily driven by thalamic disinhibition associated with secondary thalamic degeneration, rather than abnormalities localized to the cortex or white matter connecting cortex and thalamus. More broadly, these results highlight thalamic dysfunction as a potentially important determinant of poststroke outcomes worthy of further investigation, and demonstrate the efficacy of physiological models to noninvasively index thalamic pathophysiology for that purpose.

Although the neurobiological processes causing secondary degeneration after stroke are unclear, the link we identified between thalamus degeneration and thalamic disinhibition (via reticular nucleus dysfunction) appears plausible given previous work. Experimental brain infarcts in rats have been shown to cause morphological changes and neuronal loss in the reticular nucleus, as well as retrograde degeneration in the relay nuclei associated with the infarcted tissue (47, 57). Crucially, one study found the affected reticular nucleus cells to have persistent reductions in intrinsic excitability, which could in turn cause disinhibition of their associated relay cells (57). Furthermore, these abnormalities were found in reticular nucleus cells projecting to both degenerated and less affected relay nuclei, suggesting that focal cortical infarcts could cause a general disinhibition of the thalamus, consistent with our findings here. Indeed, the balance of inhibition relative to excitation has been identified as a possible determinant of power spectrum slope, and therefore potentially stroke-related spectral slowing, although the directionality remains unclear (see discussion in ref. 18). The causes of secondary reticular nucleus damage are also unknown, but work in rodents (47) and humans (42) suggests that neurotransmitter imbalance following disruption of GABAergic projections from the globus pallidus could play an important role. In sum, there is some evidence that focal infarcts can cause both secondary damage of thalamic relay cells and thalamic disinhibition

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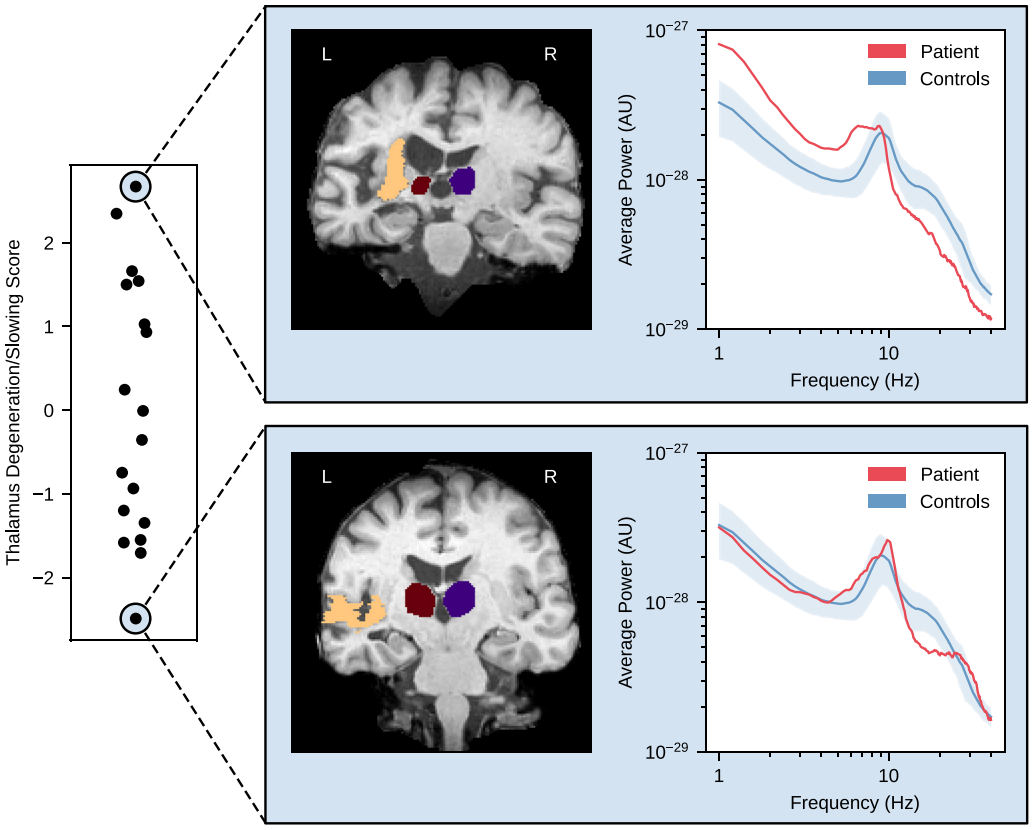


Fig. 5. Thalamic atrophy and MEG slowing in individual patients. Examples depicting a patient with a high degree of thalamus degeneration and MEG slowing (*Top*) and a patient with a low degree of thalamus degeneration and MEG slowing (*Bottom*). (*Left*) Depicts the distribution of scores on the PCA depicted in Fig. 4*B*. (*Center* column) Coronal slices, maroon = left thalamus, purple = right thalamus, yellow = lesion. (*Right* column) Average left hemisphere power spectra, comparing the given patient to controls. The shaded area represents SD of the control group.

via reticular nucleus disruption, but more work is needed to determine whether one causes the other, or whether they co-occur due to a shared cause [e.g., destruction of mutual corticothalamic axons (57)]. Future work must also consider how the distinct subnuclei of the thalamus, with their specialized functions and anatomical connectivity, may be differently impacted by secondary disruption and implicated in impairment.

While we and other authors have focused primarily on the ipsilesional thalamus, we note that some abnormalities were detected in the contralesional (right) thalamus in our sample. Similar to the ipsilesional thalamus, the fitted corticothalamic models also showed reduced contralesional thalamic inhibition in the patient group compared to controls (Section 1.2). However, we found no evidence for degeneration of the contralesional thalamus itself, and in fact found it to be slightly larger in the patient group compared to controls, on average. A possible explanation is that some degree of degeneration was present in the contralesional thalamus in some patients, but was not detectable statistically due to heterogeneity within the sample. Alternatively, the corticothalamic model fitting may have overestimated the amount of thalamic disinhibition in contralesional thalamus due to some form of contamination from the lesioned hemisphere, e.g. if the abnormal dynamics originating from the lesioned hemisphere were capable of entraining similar dynamics in the contralesional hemisphere through interhemispheric connections. At present, we cannot conclude what role the contralesional thalamus plays in poststroke cognitive function, but future work should consider whether it can be affected by the same degenerative processes affecting the ipsilesional thalamus, or whether it may in fact play a compensatory role.

We would also like to clarify that thalamus dysfunction and abnormal dynamics must still be understood in the context of the primary injury, and the extent of infarction may be more predictive of outcomes for many patients. Indeed, we found that aphasia symptoms indexed by the WAB were better predicted by primary damage to the left inferior frontal gyrus, in contrast to the BNT and PPVT which were better predicted by secondary thalamus degeneration. Furthermore, our mixed-effects model indicated that MEG slowing at the individual region level was best predicted by the region's proximity to the lesion, potentially highlighting the importance of localized tissue disruption near the infarct itself, although localized dysfunction of specific subnuclei of the thalamus (below the resolution of our segmentation) may also play a role. Despite the importance of these more direct effects of infarction, including loss of cortical tissue, we have demonstrated that thalamic dysfunction appears to be an important sequela of stroke, with broad impacts on brain dynamics and cognitive function. We therefore argue that secondary thalamic dysfunction poststroke may be a common factor contributing to poorer clinical outcomes and recovery across patients with different lesion types. More generally, our findings also highlight the need to consider how poststroke deficits may arise from otherwise overlooked secondary damage to other subcortical structures (31).

Limitations to the generalizability of this work are the sample size and lack of females in the sample (2/18 patients). Although the sample contained only 18 patients in total, these patients were more deeply characterized than most in terms of neuroimaging measures. This was valuable, as it allowed us to use the convergence of multiple measures (e.g., volume, MD, CBF) to increase

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Table 1. Mixed effects modeling results

Fixed effects	Estimate (SE)	*t*	*P*
Intercept	0.042 (0.155)	0.272	0.789
L thalamus degeneration score	0.267 (0.147)	1.808	0.090
Proximity to lesion* **	**0.362 (0.077)**	**4.740**	**<0.001**
Lesion volume	−0.162 (0.144)	−1.125	0.278
L Thal Degen: Prox to Lesion	−0.057 (0.077)	−0.743	0.467
Random effects			
Intercept (within patient)	0.393 (0.626)		
Slope (within patient)	0.066 (0.257)		
Residual	0.463 (0.681)		

Linear mixed effects modeling suggests that MEG slowing score at the level of individual ipsilesional brain regions is best predicted by proximity to the lesion, with a possible additional effect of ipsilesional thalamus degeneration (not significant at $\alpha = 0.05$). Lesion volume was not a significant predictor of MEG slowing score at the region level. Model details: the mixed effects model predicted individual MEG slowing scores for each region for each patient, and included proximity to lesion (per region × patient), left thalamus degeneration score (per patient), lesion volume (per patient), and the interaction between lesion proximity and thalamus degeneration as fixed effects. The model included a random intercept and slope for each patient. Fixed effects were estimated with bootstrap resampling (1,000 resamples). Bolded values are statistically significant (*** $P < 0.001$).

confidence in our estimates of variables relevant to poststroke dysfunction. However, an important next step will be to replicate these findings in a more representative sample, including patients with right hemisphere lesions and more diverse, nonaphasia symptoms. Regarding the sex imbalance, it is worth noting that the original sample of 23 patients contained 6 females, but 4 of these were excluded due to the discovery of lesions within the thalamus itself (compared to 1 out of 17 males). In addition to correcting the sex imbalance in recruiting, future work should consider how thalamic degeneration and frank thalamic infarction may manifest differently in males and females.

Finally, further clarifying the causes of abnormal poststroke dynamics may require multimodal approaches that consider both finer and broader scales of brain organization. First, to better understand the nature of thalamus dysfunction, future work should consider its neurophysiology and anatomy in more detail, incorporating heterogeneous subnuclei and more direct functional measures recorded from the thalamus itself across a variety of contexts, including during sleep and cognitive tasks. Such direct measurements are needed to verify the evidence for thalamic disinhibition provided here, and would ideally provide necessary details about how it manifests within thalamic microcircuits. On the other hand, a whole-brain generative model (58) incorporating both local and long-range connectivity between corticothalamic nodes (59) could better contrast the effects of thalamic dysfunction compared to abnormal network topology (arising from lesioning of nodes and edges) as a function of individual lesion morphology (60). Both approaches may benefit from physiological models incorporating plasticity, allowing their parameters to change in response to perturbations (61). Overall, these more developed models could be used to verify the current findings, test the effectiveness of simulated treatments such as brain stimulation or pharmacotherapy with respect to normalizing thalamic dysfunction, and could accelerate clinical research by providing tools for stratifying patients and calibrating treatment parameters, similar to ongoing work in epilepsy (62).

In conclusion, we demonstrate that thalamic degeneration in chronic stroke is predictive of abnormal large-scale dynamics as measured by MEG, and this link was mediated by disinhibition of the thalamus as revealed through physiological modeling of a corticothalamic circuit. We therefore argue that thalamic disruption plays an important role in poststroke brain dysfunction that warrants further investigation, and computational modeling of electrophysiological dynamics has an important role to play in this investigation.

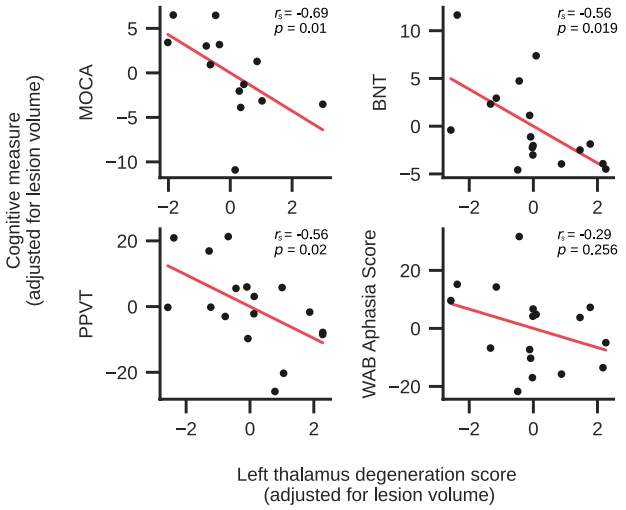


Fig. 6. Thalamus degeneration predicts cognitive and language impairment. After correcting for lesion volume, higher ipsilesional (left) thalamus degeneration scores are associated with general cognitive impairment as measured by the MOCA (n = 13), as well as worse performance on the BNT (n = 17), PPVT (n = 17), and WAB (n = 17, not significant at $\alpha < 0.05$, two-tailed). Reported statistics are Spearman correlations computed on the residuals after regressing both independent and dependent variables on lesion volume.

3. Materials and Methods

3.1. Participants. A sample of chronic stroke patients was drawn from two prior studies on poststroke aphasia collected at the same site (9, 63). Of the 23 patients available, 5 patients (4 females) were excluded after thalamic lesions were detected on their anatomical scans, as we aimed to investigate the effects of secondary thalamic degeneration, rather than the effect of primary infarction of the thalamus itself. The final sample included 18 patients (16 males), with an average age of 61 y (SD = 13) and 16.9 y of education (SD = 3.4). All patients had a single left-hemisphere stroke at least 7 mo prior to data collection (average time post onset = 4 y, SD = 3 y 4 mo) and demonstrated symptoms of aphasia. All patients were right-handed. See Table 3 for further details about patient demographics.

Both cohorts completed the WAB (64), BNT (65), and PPVT (66) to quantify their aphasia symptoms. The MOCA (67) was completed by 13 patients and was also considered here as a measure of general cognitive impairment.

Patients were matched with a sample of 23 healthy control participants (17 males, 6 females) for age ($t(39) = -1.62$, $P = 0.11$) and education ($t(39) = -0.77$, $P = 0.44$). Control participants had a mean age of 66.9 y (SD = 10.4) and 17.7 y of education (SD = 2.4).

All data collection took place at the Rotman Research Institute in Toronto, Canada, and the study protocol was approved by the Research Ethics Board at Baycrest Health Sciences. All participants gave written informed consent and were compensated for their participation. For further details about recruitment and exclusion criteria, see refs. 9 and 63.

3.2. MEG Collection and Preprocessing. Resting-state MEG data were acquired with a 151-channel CTF MEG system with axial-gradiometers and synthetic third-order gradient noise reduction applied (68). Recordings were collected in a seated position for 5 min at a sampling rate of 625 Hz (14 patients, 19 controls) (9) or for 10 min at 1,250 Hz (4 patients, 4 controls) (63), while participants fixated on a white cross presented on a black background. Continuous MEG datasets were downsampled to 625 Hz (if collected at 1,250 Hz), band-pass filtered between 0.5 and 125 Hz, and divided into 5-s epochs. Bad channels and epochs contaminated by large artifacts were manually flagged and excluded from further analysis.

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Table 2. Thalamus degeneration and primary tissue damage differentially predict language impairment

	BNT		PPVT		WAB	
n (df)	17 (12)		17 (12)		17 (12)	
F statistic (p)	2.25 (0.124)		2.36 (0.111)		7.04 (0.004)**	
Log-Likelihood	−19.37		−19.18		−13.85	
AIC	48.73		48.36		37.71	
BIC	52.90		52.53		41.87	
	Coef (SE)	t (*P*)	Coef (SE)	t (*P*)	Coef (SE)	t (*P*)
Constant	−0.11 (0.23)	−0.49 (0.633)	0.026 (0.22)	0.12 (0.907)	−0.15 (0.16)	−0.92 (0.374)
L Thal Degen Score	**−0.55 (0.23)**	**−2.34 (0.035)***	**−0.51 (0.23)**	**−2.22 (0.047)***	−0.06 (0.17)	−0.36 (0.727)
L Ang Gyr Damage	−0.17 (0.27)	−0.63 (0.543)	0.03 (0.24)	0.128 (0.900)	0.17 (0.19)	0.86 (0.408)
L Sup Temp Gyr Damage	−0.01 (0.29)	−0.05 (0.962)	−0.28 (0.28)	−0.97 (0.351)	−0.25 (0.21)	−1.19 (0.258)
L Inf Front Gyr Damage	−0.38 (0.24)	−1.135 (0.279)	−0.17 (0.27)	−0.65 (0.526)	**−0.90 (0.24)**	**−3.70 (0.003)****

Multiple linear regression indicates that ipsilesional (left) thalamus degeneration score significantly predicts BNT and PPVT performance, while the WAB was better predicted by primary damage to the left inferior frontal gyrus. Bolded values are statistically significant (* $P < 0.05$, ** $P < 0.01$).

To localize signals to specific brain regions, the AAL atlas (52) was first nonlinearly registered from MNI space to each subject's native space T1 image using ANTS (69). Synthetic aperture magnetometry (SAM) beamforming (68) was then used to localize signals to virtual channels centered on the 78 cortical regions of interest specified by the atlas. Specifically, beamforming target voxels were chosen by placing spheres 10 voxels (10 mm) in diameter at the center of mass of each region. After computing a head model from the T1 image aligned to the MEG coordinate system, CTF software (CTF; Port Coquitlam, British Columbia, Canada) was used to generate a set of beamformer weights (spatial filters) for each voxel within each target sphere. To summarize the activity across the voxels in a given sphere, singular value decomposition (SVD) was computed on the matrix of beamformer weights associated with each sphere in MATLAB, and the first singular value was multiplied by the sensor time series for each epoch, producing a single 5 s time series for each sphere (hereafter "virtual channel") at each epoch (70). Any target voxels that overlapped with the lesion mask were excluded from the SVD, and spheres that overlapped >50% with the lesion mask were excluded entirely from further analyses. In the ipsilesional hemisphere, this left virtual channels intact in 628 regions across all patients, or an average of 34.9 out of 39 left hemisphere regions per patient. Further MEG preprocessing details are reported in ref. 18.

3.3. Quantifying MEG Spectral Abnormalities. Stroke-related MEG abnormalities in the ipsilesional hemisphere were quantified by first computing the power spectral density (PSD) at each of the 78 virtual channels for each epoch [NeuroDSP (71); "medfilt" smoothed FFT], then averaging these PSDs over all good epochs. We then applied Specparam (72) to the PSD to parameterize the power spectrum and disambiguate aperiodic from periodic features (1 to 50 Hz, no knee, maximum four peaks).

Since increased aperiodic exponent, decreased alpha frequency, and decreased beta power have been shown together to index dysfunction (18), these three parameters were combined into a single "MEG slowing score" using PCA. Because stroke has both broadly distributed and localized effects in the ipsilesional hemisphere, MEG slowing scores were computed for patients 1) across the entire left (ipsilesional) hemisphere by averaging over all left hemisphere cortical regions, and 2)

Table 3. Patient demographic and clinical information

Patient	Age (years)	Education (years)	Sex	Handedness	Time post onset	Etiology	Lesion volume (mm^3)	% brain volume lesioned	Cohort
P1	47	18	M	R	4 y 1 mo	Ischemic	18,656	1.37	K
P2	67	21	M	R	15 y 6 mo	Hemorrhagic	217,065	13.76	K
P3	70	24	M	R	1 y	Ischemic	2,952	0.21	K
P4	75	15	M	R	2 y 5 mo	Ischemic	48,137	2.54	K
P5	79	10	M	R	2 y 1 mo	Ischemic	32,901	2.44	K
P6	46	15	M	R	2 y 3 mo	Ischemic	30,861	2.2	K
P7	62	16	M	R	1 y 2 mo	Ischemic	38,064	2.61	K
P8	77	20	M	R	7 mo	Ischemic	22,910	1.38	K
P9	66	20	M	R	5 y 3 mo	Unspecified	81,473	5.86	K
P10	46	16	M	R	4 y	Ischemic	99,098	6.71	K
P11	57	12	M	R	2 y	Unspecified	115,811	8.81	K
P12	65	20	M	R	7 y 1 mo	Ischemic	177,022	12.72	K
P13	69	15	M	R	1 y	Ischemic	7,645	0.57	K
P14	68	14	F	R	5 y 6 mo	Hemorrhagic	4,5874	3.24	K
P15	61	16	M	R	4 y	Ischemic	27,118	2.05	SB
P16	34	19	F	R	4 y	Ischemic	21,004	1.55	SB
P17	41	18	M	R	5 y 9 mo	Ischemic	50,942	3.9	SB
P18	68	16	M	R	4 y 10 mo	Hemorrhagic	10,165	0.67	SB
Mean	61.00	16.94			4 y		58,205.44	4.03	
SD	13.03	3.40			3 y 4 mo		59,440.09	4.03	

K = Kielar cohort (9), SB = Shah-Basak cohort (63).

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for individual cortical regions within the left hemisphere. In both cases spectral parameters were normalized by subtracting the mean and dividing by the SD of the corresponding region/hemisphere in controls prior to PCA. Only 447 of the 628 unlesioned left hemisphere cortical virtual channels had both an alpha and beta peak detectable by Specparam, so the remainder were excluded from the analysis computed over individual cortical regions.

The first principal component derived from these variables captured the MEG slowing phenomenon as previously described (18) in both the whole hemisphere averages (PC1 loadings: exponent = 0.63, alpha freq = −0.67, beta power = −0.38; variance explained = 55.2%) as well as at individual regions (PC1 loadings: exponent = 0.51, alpha freq = −0.73, beta power = −0.45; variance explained = 43.7%). Projections onto these principal components were considered MEG slowing scores for a given patient/region in subsequent analyses.

3.4. Corticothalamic Model. The corticothalamic model emulates the dynamics of a corticothalamic circuit comprising four coupled neuronal populations: 1) excitatory cells (*e*) and 2) inhibitory cells (*i*) in a patch of the cortex, 3) cells in the excitatory relay nuclei of the thalamus (*s*), and 4) cells in the inhibitory reticular nucleus of the thalamus (*r*).

SI Appendix describes in detail the second-order differential equations that define these populations' aggregate membrane potential and firing rate dynamics, as well as the linearized frequency domain variant used to analytically derive the power spectra used here for model fitting.

Parameter estimation was carried out with BrainTrak, a software library designed to efficiently fit the linearized frequency domain model to empirical PSD data (24, 51, 73). The model parameters of primary interest were based on the gain parameters G_{ab}, which reflect the magnitude of response in population *a* relative to the magnitude of input from population *b*:

$$G_{ab} = \rho_a v_{ab} = \rho_a N_{ab} s_{ab},$$

where ρ_a is the slope of the linear approximation to population *a*'s sigmoidal wave-to-pulse firing rate function, N_{ab} is the mean number of synapses from neurons of population *b* to population *a*, and s_{ab} is the mean strength of the synapses. We estimated the gains G_{ee} and G_{ei}, as well as three composite gain parameters defined as $G_{ese} = G_{es}G_{se}$, $G_{esre} = G_{es}G_{sr}G_{re}$ and $G_{srs} = G_{sr}G_{rs}$. The composite gains G_{ese} and G_{esre} represent the overall gain within the two loops traversing the cortex and thalamus, while G_{srs} represents the gain within the intrathalamic loop comprising the relay nuclei and the reticular nucleus (Fig. 1*A*). The three gains involving an inhibitory population (G_{ei}, G_{esre}, and G_{srs}) are negative, while G_{ee} and G_{ese} are positive. In addition to the gains, we fit the inverse synaptodendritic decay time α, the inverse synaptodendritic rise time β, the thalamocortical time delay t_0, and the amplitude of the electromyographic artifact correction A_{EMG}.

For ease of interpretation, the gain parameters were further aggregated into three additional parameters summarizing the overall gain within the three large-scale loops of the model: *X* (intracortical loop gain), *Y* (corticothalamic loop gain), and *Z* (intrathalamic loop gain):

$$X = \frac{G_{ee}}{1 - G_{ei}},$$

$$Y = \frac{G_{ese} + G_{esre}}{(1 - G_{srs})(1 - G_{ei})},$$

$$Z = -G_{srs}\frac{\alpha\beta}{(\alpha + \beta)^2}.$$

For model fitting, BrainTrak implements a Metropolis–Hastings MCMC algorithm to sample the joint probability distribution of the model parameters and estimate the parameter combination that produces maximum goodness-of-fit. See *SI Appendix* and ref. 51 for further details about parameter estimation in BrainTrak.

We applied BrainTrak to fit the average PSDs at each of the 78 cortical regions for each participant, excluding any virtual channels that were more than 50% lesioned as noted above. The fitting range was restricted to 1 to 40 Hz, with no spatial variation in model parameters, and an MCMC chain length of 100,000 was used for sampling the joint probability distribution.

3.5. MRI Acquisition and Processing. All MRI images were acquired with a Siemens 3T TIM Trio located at the Rotman Research Institute, with all sequences collected during the same session. All DICOMS were converted to NIFTI with Heudiconv (74) v0.11.6.

3.5.1. Structural imaging. A T1-weighted anatomical image was collected for all 18 patients and 23 controls [MPRAGE, 1 mm isotropic voxels, repetition time (TR) = 2 s, echo time (TE) = 2.63 ms, FOV = 256 × 256 mm^2, 160 axial slices]. Additionally, a T2-FLAIR image (0.9 × 0.9 × 5.0 mm voxels, TR = 9 s, TE = 96 ms, FOV = 224 × 224 mm, 32 axial slices) was collected and coregistered with the T1 for lesion segmentation in patients (see ref. 9 for more information about lesion segmentation).

3.5.2. MD quantification. MD was estimated for 17 patients and 23 controls using diffusion-weighted images acquired with 30 gradient directions (b = 1,000 s/mm^2) and one reference image (b = 0 s/mm^2). EPI parameters were as follows: 2.2 mm isotropic voxels, 110 × 110 matrix, 68 axial slices, TR = 7.9 s, TE = 84 ms, FOV = 242 × 242 mm.

To fit diffusion tensors and thereby estimate MD, a brain mask was first extracted from the reference image using FSL's (75) Brain Extraction Tool (76), then eddy was applied to the brain-masked diffusion-weighted images for outlier removal, motion correction, and eddy current correction [Eddy CUDA 9.1 (77–79); FWHM = 2, 16 degrees of freedom]. Finally, diffusion tensors were fitted with dtifit (80) using the rotated b vectors generated from eddy, and the resulting MD map was registered to the anatomical T1 image using a rigid transform computed on the reference image with epi_reg (FSL v6.0.4).

3.5.3. CBF quantification. CBF was estimated for 16 patients and 23 controls using a PICORE Q2TIPS pulsed ASL sequence (81) (pASL) with an inversion time of 1,800 ms (TI_1 = 700 ms, TI_{1S} = 1,600 ms, TI_2 = 1,800 ms), 75 control-tag pairs, and a flow limit of 100 cm/s. The labeling region was 110 mm thick, with a 29 mm gap between labeling and imaging regions. GRAPPA EPI readout parameters were as follows: 3.5 × 3.5 ×5 mm voxels, 0.25 mm between slices, 64 × 64 matrix, 26 axial slices, acceleration factor = 2, TR = 3 s, TE = 13 ms, and field of view (FOV) = 224 × 224 mm. A calibration image (m_0) was acquired with a 2D gradient-echo EPI scan with a TR of 10 s.

Bad control-tag pairs were identified with the SCORE algorithm as implemented by ASLprep (82, 83) (v0.2.7), and excluded from further processing. BASIL was applied to the remaining control-tag pairs for motion correction, control-tag subtraction, voxelwise calibration to the m_0 image, and finally registration and interpolation to the anatomical T1 image space [oxford_asl (84); FSL v6.0.4].

3.6. Quantifying Thalamic Degeneration. The spatial extents of the left and right thalami were first approximated by applying Freesurfer (85) (recon-all) to the T1 image. Binary thalamus masks were then created by extracting the left and right thalami from Freesurfer's volumetric subcortical segmentation. These masks were manually corrected by PRJ following a published thalamus segmentation protocol (54) using the T1 image, inverted T1 image, and coregistered FA image to guide segmentation with ITK-Snap (86). All manual segmentations were then verified by a second rater (LR). As previously noted, any patients with detectable lesions within either the left or right thalamus were excluded from all analyses.

Ipsilesional thalamic degeneration was assessed using a combination of left thalamus volume, MD, and CBF, which have individually been associated with secondary poststroke degeneration in prior work. Thalamus volume (31–33, 36, 37) was computed simply as the number of voxels (1 mm isometric) within the binary thalamus mask. MD (34, 38–41) and CBF (42–44) values were computed by multiplying the binary thalamus masks with the T1-registered MD and calibrated CBF maps, respectively, then computing the mean. Finally, each volume, MD, and CBF estimate was normalized by subtracting the mean and dividing by the SD of the left thalamus in controls. The same procedure was repeated for the contralesional (right) thalamus, and these values were normalized by the right thalamus in controls.

3.7. Statistical Analysis. Group differences between patients and controls were assessed with independent samples t-tests, with a two-tailed significance threshold of $\alpha = 0.05$ before Bonferroni correction for multiple comparisons, where applicable. When considering the stroke group alone, all raw variables (volume, MD, CBF, aperiodic exponent, alpha peak, beta power) were standardized by the control group's mean and SD at the given region/hemisphere prior to subsequent analysis. Linear relationships between variables within the patient group were assessed with linear regression [Statsmodels v0.14.4 (87)]. PCA was used to reduce multivariate patterns among variables into a single score for each patient (e.g., to produce a single metric of thalamic degeneration or spectral slowing; see Section 1 for details). Three missing values (1/18 patient MD estimates and

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2/18 patient CBF estimates) were replaced with the mean of the patient group, and all values were Z-scored prior to PCA. Causal Mediation Analysis was used to test whether model-estimated thalamic disinhibition mediated the relationship between thalamus degeneration and MEG slowing (55) (Section 1.4). Finally, the effects of thalamic degeneration versus proximity to the lesion itself on the power spectrum were assessed with a mixed effects model (56) (Section 1.5).

Data, Materials, and Software Availability. Preprocessed data and analysis code are provided at https://github.com/McIntosh-Lab/stroke_ct_model (88). Raw data are available at http://www.braincode.ca/content/controlled-data-releases#dr023 (89). Previously published data were used for this work (9, 63).

ACKNOWLEDGMENTS. We thank Aneta Kielar and Priyanka Shah-Basak for data collection, and Taha Morshedzadeh, Sorenza Bastiaens, Kevin Kadak, and Zheng Wang for helpful discussions about the corticothalamic model. This research was enabled by computing resources provided by Calcul Québec (calculquebec.ca) and the Digital Research Alliance of Canada (alliancecan.ca). P.R.J.: Natural Sciences and Engineering Research Council of Canada (NSERC) CGS-D; J.D.G.: Krembil Foundation, CAMH Discovery Fund, Labatt Family Network, UToronto EMHSeed, A.R.M.: NSERC Discovery Grant; J.A.M.: Canada Research Chair, Canadian Partnership for Stroke Recovery Catalyst Grant, NSERC Discovery Grant.

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